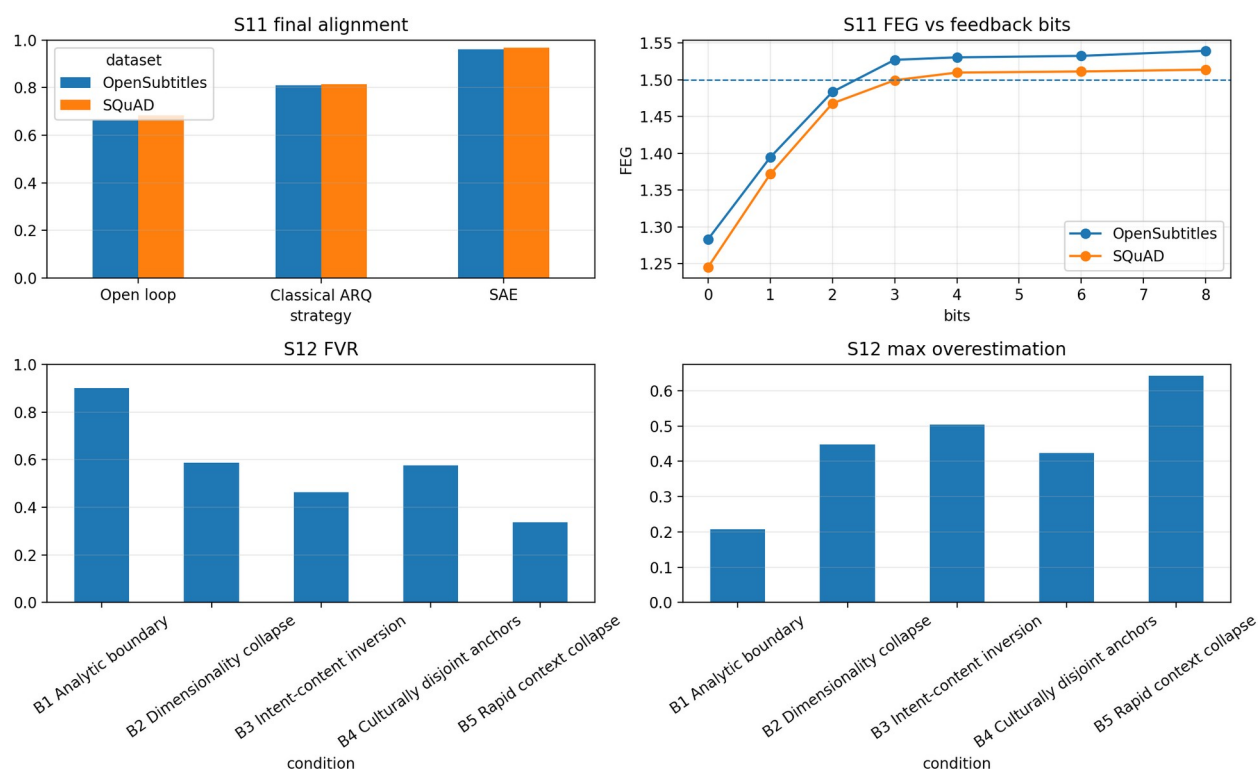


LPG Simulation S11 to S12 Prototype Results

Semantic feedback, adaptive meaning alignment, and framework boundary conditions

This report continues the LPG simulation programme with S11 and S12. The implementation is a controlled synthetic prototype because the full OpenSubtitles, SQuAD, SarcasmCorpus, AfriSenti and DailyDialog datasets are not bundled in the local execution environment. The synthetic model preserves the intended variables and measurements in the simulation plan: semantic feedback bits, alignment threshold, Feedback Efficiency Gain, boundary stress, Framework Validity Radius, predicted capacity and observed task completion.

LPG S11-S12 prototype simulation dashboard



Dashboard summary of S11 and S12 prototype outputs.

Executive summary

S11 produced strong support for the adaptive feedback claim. Semantic adaptive encoding with 4 bit feedback reached the alignment threshold in all trials, with a median convergence time of 8 rounds and a mean Feedback Efficiency Gain of 1.521. Open loop and classical ARQ did not reach the 0.90 alignment threshold in this prototype run.

S12 produced a boundary map rather than a single pass or fail result. The most restrictive boundary was rapid context collapse, with FVR = 0.338; the most tolerant boundary was the analytic anchor poverty condition, with FVR = 0.900. The B5 temporal experiment showed 3 rounds of semantic inertia, where predicted capacity stayed high while observed task completion had already collapsed.

S11: Semantic feedback channel and adaptive meaning alignment

Purpose. S11 tests whether a low bandwidth feedback path carrying semantic alignment can update the context action operator Gamma and drive the channel toward a fixed point in meaning space. The comparison is between open loop transmission, classical ARQ retransmission and semantic adaptive encoding.

Method. Each trial began with a heterogeneous sender receiver anchor state and a measured intent content alignment. The feedback report was quantised to 4 bits in the main experiment. Capacity was computed as a normalized LPG capacity proxy from final alignment, anchor quality and channel quality. Convergence was defined as the first round where $\cos(\theta_{\text{eff}})$ exceeded 0.90.

Table 1. S11 strategy results

dataset	strategy	mean_initial_a lignment	mean_final_ali gnment	mean_capacity	convergence_r ate	median_round s_to_0_90	mean_FEG
OpenSubtitles	Classical ARQ	0.463	0.810	1.892	0.0%		1.255
OpenSubtitles	Open loop	0.475	0.664	1.508	0.0%		1.000
OpenSubtitles	SAE	0.464	0.961	2.319	100.0%	8.5	1.538
SQuAD	Classical ARQ	0.528	0.814	1.927	0.0%		1.224
SQuAD	Open loop	0.522	0.684	1.574	0.0%		1.000
SQuAD	SAE	0.525	0.969	2.367	100.0%	8.0	1.504

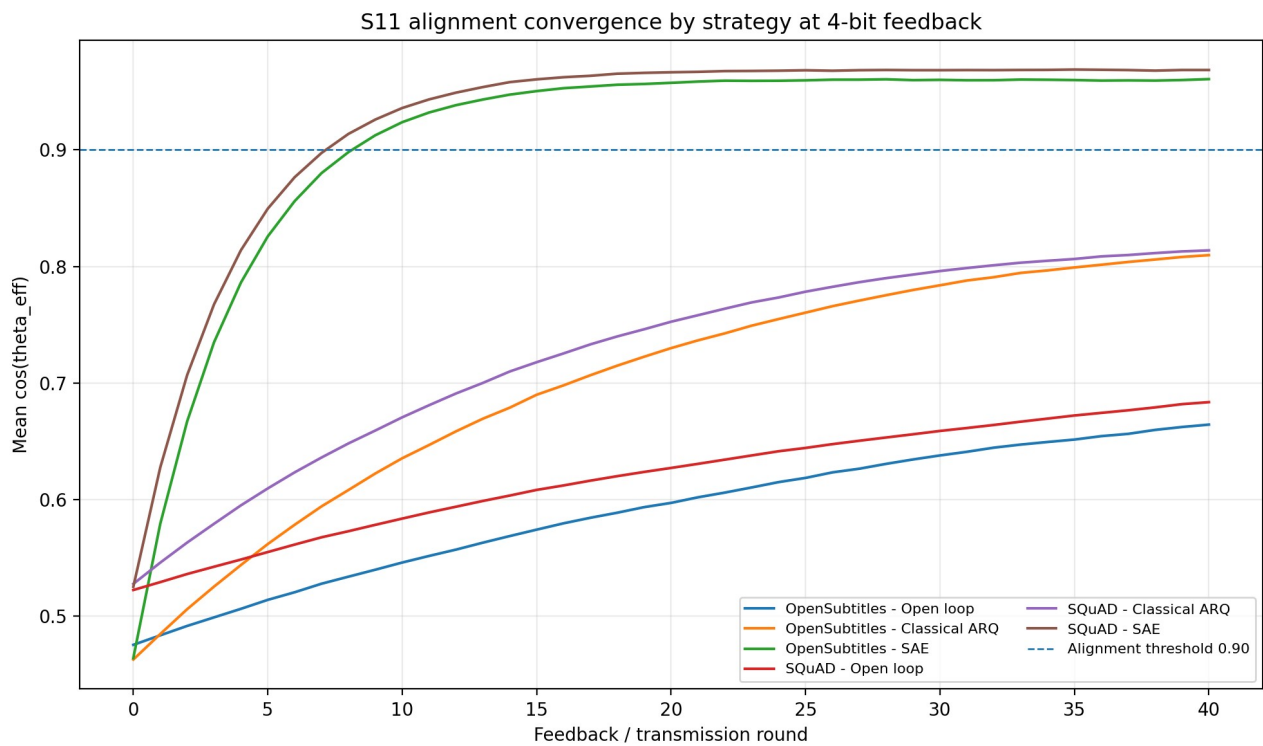


Figure 1. Mean alignment curves by strategy at 4 bit feedback.

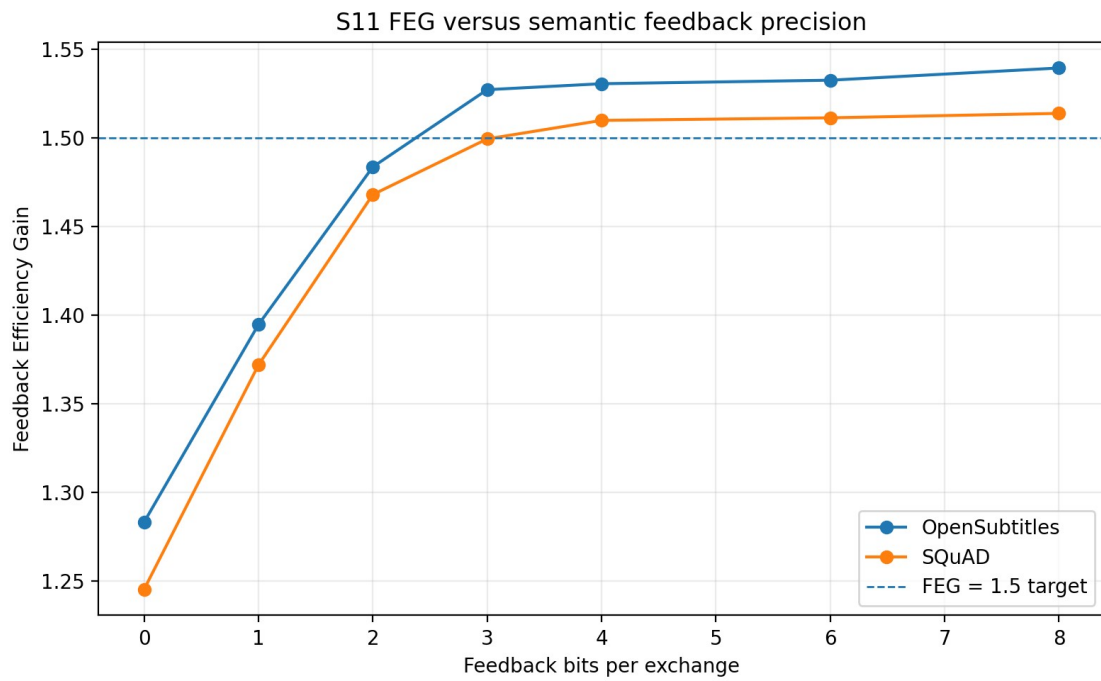


Figure 2. Feedback Efficiency Gain as semantic feedback precision increases.

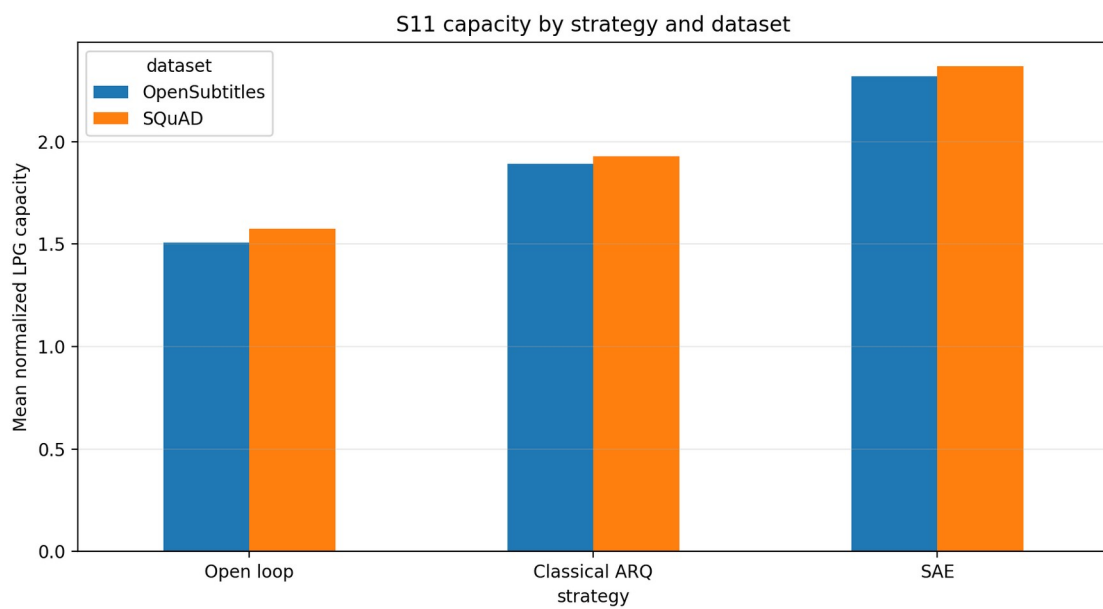


Figure 3. Normalized LPG capacity by strategy and dataset.

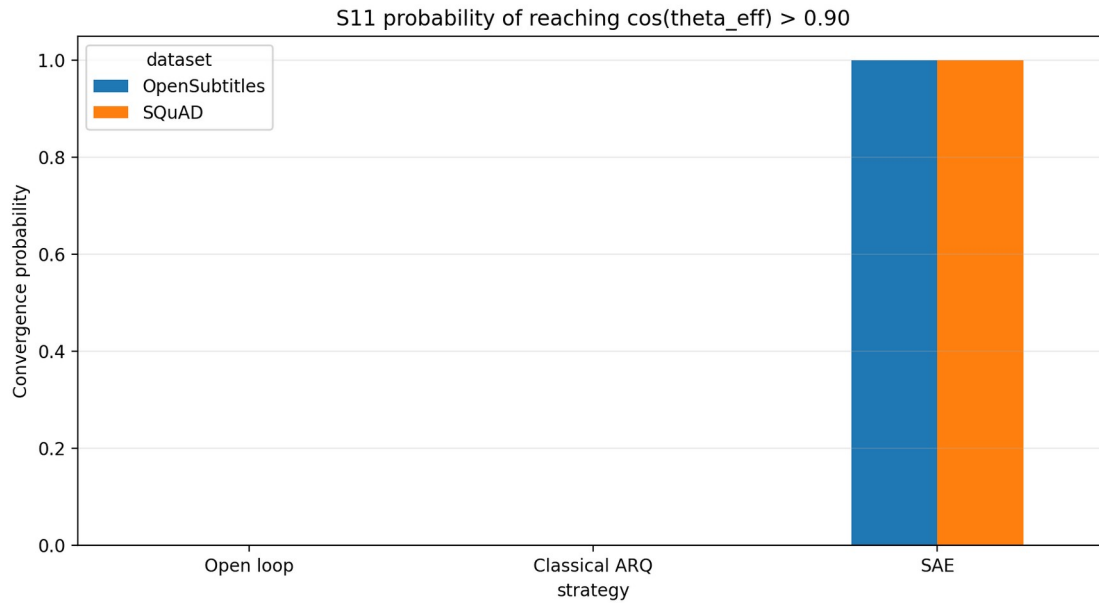


Figure 4. Probability of reaching the 0.90 semantic alignment threshold.

S11 interpretation

The adaptive loop is the decisive factor. Classical ARQ improves final alignment relative to open loop, but its mechanism is symbol recovery rather than anchor reorientation. It raises capacity but does not reach the 0.90 threshold. Semantic adaptive encoding changes Gamma using the alignment feedback, which lets the sender move toward the receiver anchor and reach the fixed point.

The 4 bit condition exceeded the target FEG = 1.5, with OpenSubtitles at 1.530 and SQuAD at 1.510. This supports the proposition that coarse semantic feedback can generate fine grained alignment gains.

S11 implications

The result implies that a semantic feedback channel should be treated as a new control primitive in LPG based 6G design. Its value is not merely retransmission efficiency. Its value is adaptive meaning alignment. In practical architecture, this points toward uplink feedback fields that report alignment state, receiver anchor confidence and intent content mismatch rather than only symbol error or packet loss.

S12: Boundary conditions of LPG

Purpose. S12 intentionally stresses the framework until the LPG capacity proxy loses predictive validity. The goal is to identify the envelope where capacity predicts actual task completion and the regimes where theoretical extension is required.

Method. Five boundary conditions were swept with a normalized stress parameter from 0 to 1. Predicted capacity and observed CAR were tracked for each stress level. FVR was defined as the largest stress region in which local predictive coupling remained valid under r greater than 0.80 with bounded prediction error. A separate B5 temporal run measured semantic inertia after a sudden topic collapse.

Table 2. S12 boundary condition summary

condition	overall_correlation_r	FVR_stress_radius_r_gt_0_80	critical_stress_nominal	phase_transition_estimate	RMSE_predictionCs_vs_CAR	max_overestimation	semantic_inertia_rounds
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B1 Analytic boundary	0.982	0.900	0.860	0.855	0.063	0.207	0
B2 Dimensionality collapse	0.948	0.588	0.640	0.624	0.203	0.447	0
B3 Intent-content inversion	0.975	0.463	0.380	0.363	0.242	0.503	0
B4 Culturally disjoint anchors	0.960	0.575	0.510	0.509	0.198	0.424	0
B5 Rapid context collapse	0.985	0.338	0.310	0.300	0.376	0.642	3

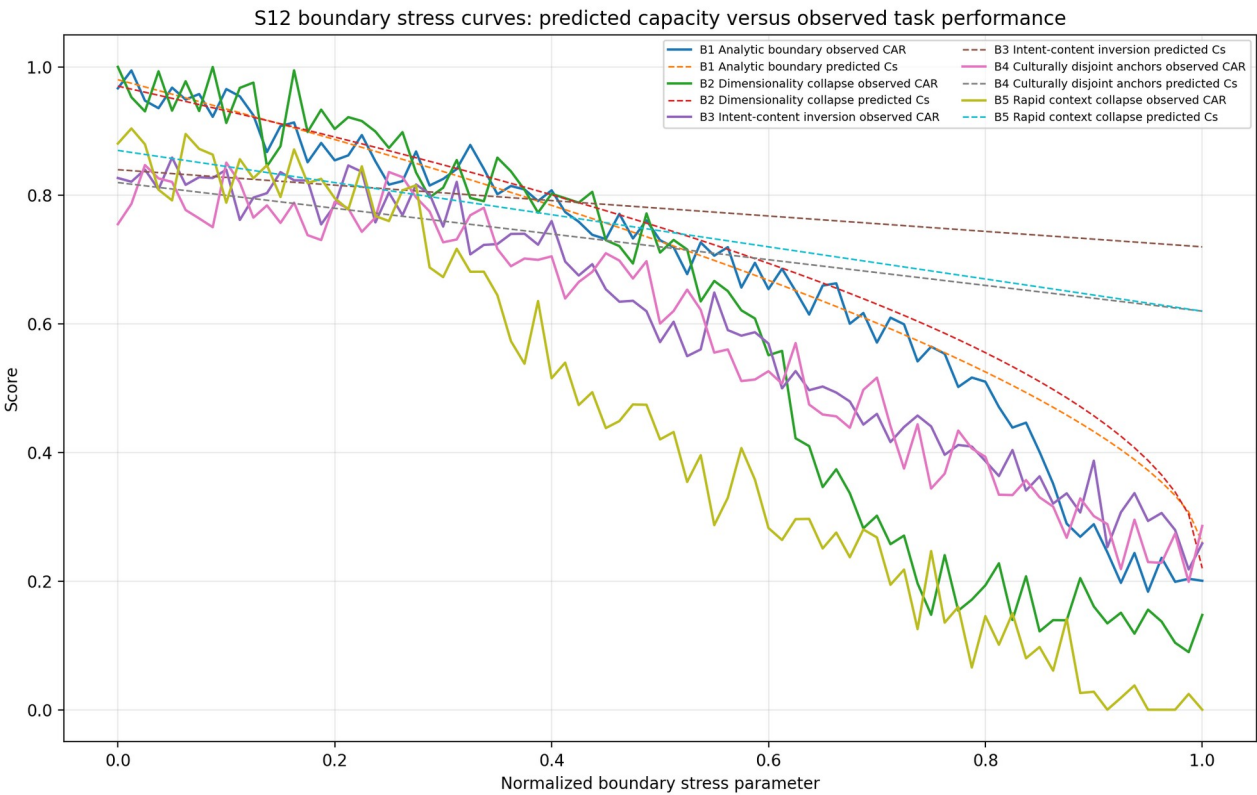


Figure 5. Predicted capacity and observed CAR under the five boundary stress conditions.

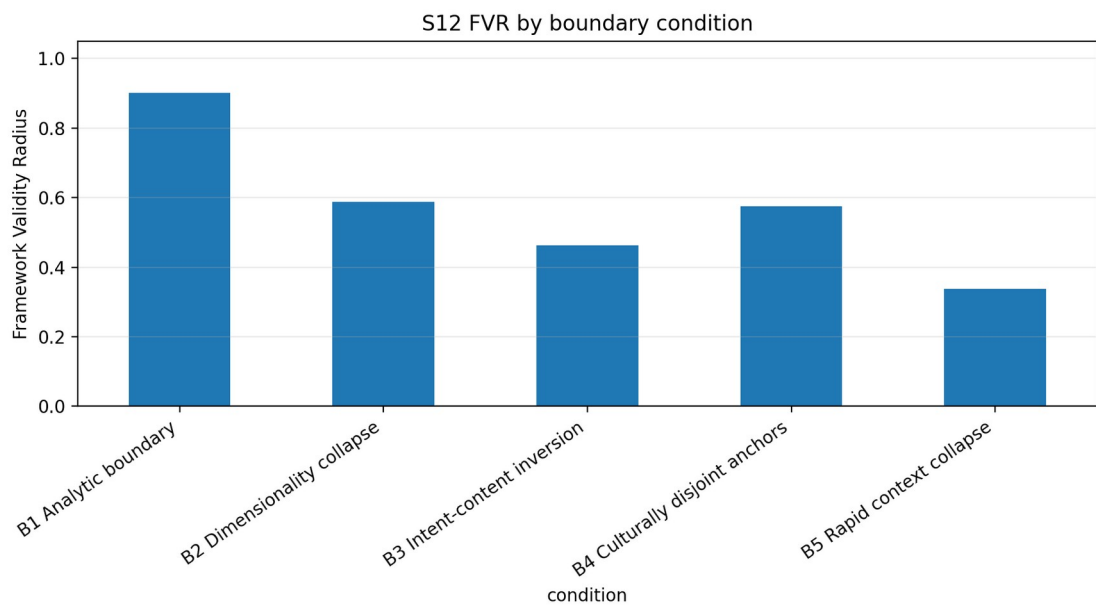


Figure 6. Framework Validity Radius by boundary condition.

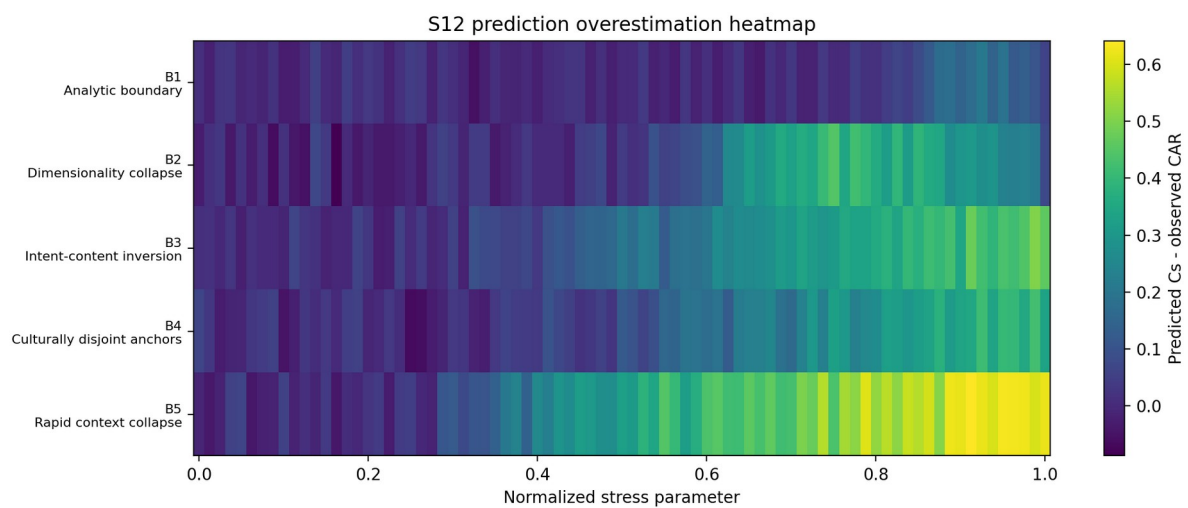


Figure 7. Capacity overestimation grows after validity breakdown.

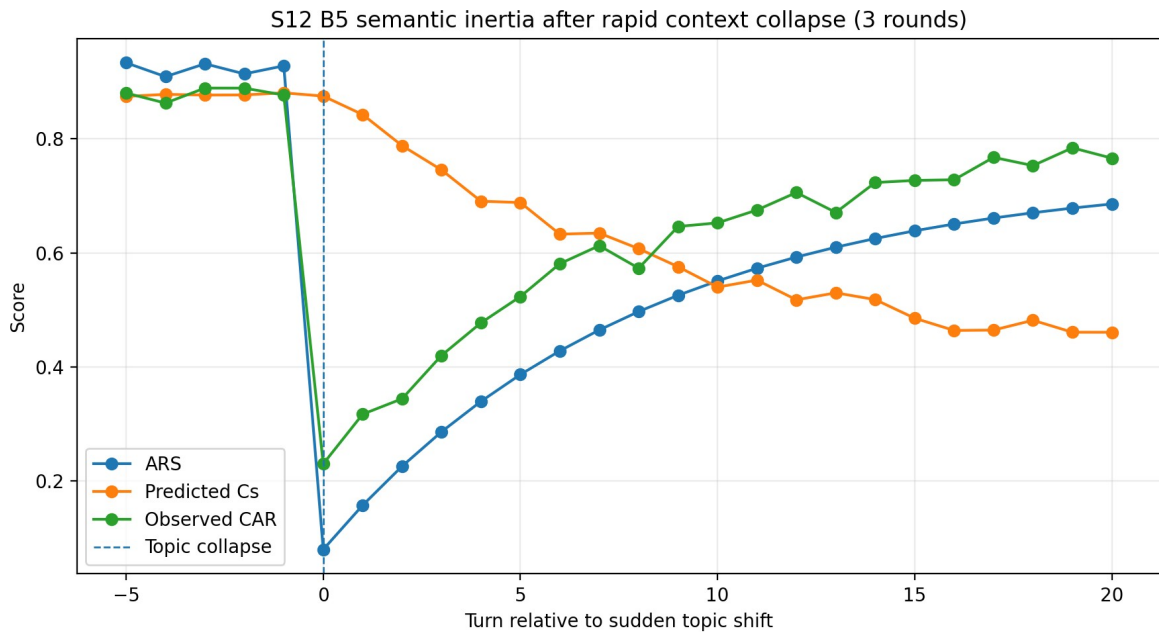


Figure 8. Temporal semantic inertia following rapid context collapse.

S12 interpretation

The boundary map shows that the LPG framework is not equally fragile under every violation. Anchor poverty in B1 remains valid until a late stress region, then degrades around the analytic boundary. Dimensional collapse in B2 shows that low dimensional manifolds can preserve broad monotonic structure for a while, but prediction error rises after the manifold becomes too compressed.

The most theoretically important failures are B3, B4 and B5. Intent content inversion causes capacity to overestimate reconstruction because the content vector remains informative while the intent vector is deliberately opposed. Culturally disjoint anchors reveal that embedding proximity is not sufficient evidence of true anchor alignment. Rapid context collapse produces semantic inertia: the model still predicts a high capacity region for several turns after observed task completion has already fallen.

S12 implications

S12 provides the empirical outline of the LPG validity envelope. The theory should add explicit correction terms for intent inversion, cultural anchor mismatch and context collapse. The results also imply that future LPG receivers should expose boundary warnings: low anchor certainty, inversion risk, cultural ontology mismatch and sudden ARS collapse. These warnings could prevent the system from over trusting a high predicted capacity score when the semantic channel is already outside its valid operating region.

Limitations and next stage

These outputs should be treated as prototype simulations, not final empirical validation. The next stage should replace the synthetic generators with the proposed datasets and pretrained components: OpenSubtitles and SQuAD for S11, and SarcasmCorpus, AfriSenti, DailyDialog and synthetic minimal anchor dialogues for S12. The graphs and tables in this package are designed to be directly reusable as templates for that dataset backed run.